6. Implementation Plan

Project: Advanced Personal Portfolio Website + Admin Panel **Builder:** Google AI Studio (Build / Max mode) **Version:** 1.0

> **Roman Urdu summary:** Yeh document batata hai ke website kis tarteeb se banani hai, har phase ke baad kya check karna hai, AI Studio ko kaunsa prompt dena hai, aur dummy data se real data par kaise switch karna hai.

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1. How to Use the 6 Documents in AI Studio

1. Start a new Build session. Upload/paste **all six documents** (PRD, TRD, App Flow, UI/UX Brief, Backend Schema, this plan) and the original spec file.
2. Paste the **Kickoff Prompt** (§2) first. Then give **one phase prompt at a time** (§4). Do not ask for everything in one message.
3. After every phase: run the phase checklist, fix bugs in the same session, and only then continue.
4. If the builder drifts (new colours, new stack, fake data), reply: *"Re-read the UI/UX Brief and TRD and correct this to match."*
5. Keep each phase's working version (export to GitHub or download the project) before the next phase.

2. Kickoff Prompt (paste first)

```
You are a senior product designer, full-stack engineer, accessibility
specialist and performance engineer.
Build my personal portfolio website and an advanced admin panel strictly
according to the attached documents:
PRD, TRD, App Flow, UI/UX Design Brief, Backend Schema and Implementation
Plan.

Rules:
- Stack: React 18 + TypeScript + Vite + Tailwind + Motion + React Router +
TanStack Query + React Hook Form + Zod. Supabase for backend (see TRD). Do
not change the stack.
- The app must run immediately in DEMO MODE with bundled sample data
(SeedProvider) and switch to Supabase automatically when env variables exist.
- Every content row carries is_sample. Seed realistic but generic sample data
(no real brands, people, quotes, links or statistics).
- No inert buttons. If a provider key is missing, show exactly what must be
configured; never pretend a feature works.
- Follow design tokens, motion timings and accessibility rules from the UI/UX
Brief.
- Work in phases. After each phase, list what works, what needs
configuration, and known issues.
Start with Phase 1 only when I send it.
```

3. Phase Overview

Phase	Goal	Main output
0	Setup	Repo, tooling, env validation, tokens, seed JSON, data provider interface
1	Foundation	Layout, theme, navbar, footer, Home skeleton with sample content, Seo component
2	Portfolio system	Skills, projects gallery, filters/search, case-study template, resume, CV
3	Cinematic intro and motion	Intro overlay + state machine, transition, reveal, global animations, reduced motion

Phase	Goal	Main output
4	Business features	Services, pricing, FAQ, testimonials, contact form + pipeline, analytics events
5	Admin panel (core)	Auth, shell, dashboard, projects, media, settings, intro manager, inbox
6	Admin panel (advanced)	Remaining modules, sample-data manager, revisions, audit log, users, backup, analytics
7	Personalization and extras	Command palette, blog, language switcher (if ready), custom cursor/3D (optional)
8	Hardening and launch	SEO, performance, accessibility, security, tests, deployment, real data entry

4. Phase Details and Prompts

Phase 0 — Setup

Tasks: project scaffold; `config/env.ts` (Zod-validated); `config/tokens`, `motion.ts`, `flags.ts`; global CSS variables for dark/light; folder structure from TRD §4; `DataProvider` interface + `SeedProvider`; seed JSON files per Backend Schema §13; `.env.example`; `/styleguide` dev route. **Exit criteria:** app boots with zero config; sample data loads; tokens render correctly in a styleguide page. **Prompt:** *"Phase 0: scaffold the project exactly as in TRD §4, implement tokens and the DataProvider with SeedProvider and all seed JSON, add the dev-only /styleguide route. Do not build pages yet."*

Phase 1 — Foundation

Tasks: app shell, router, error boundaries; theme provider with no-flash script; glass navbar (desktop + mobile drawer with focus trap, Escape, scroll-spy, progress line); footer; Home with hero, about, skills preview using sample data; Seo component; custom 404. **Exit criteria:** every nav link works; mobile menu keyboard accessible; theme persists; no layout shift at 360, 768, 1440 px. **Prompt:** *"Phase 1: build the foundation per UI/UX Brief §3–§5.3–5.5 and App Flow V3 and V9. Hero must fit in the first viewport on a 360×640 phone."*

Phase 2 — Portfolio system

Tasks: featured projects, `/projects` with chips, debounced search, sort, URL query params, load more, empty state; `/projects/:slug` template with gallery lightbox, share, previous/next, related; skills explorer and tools wall; experience timeline; `/resume` with print CSS and CV button (only when PDF exists). **Exit criteria:** filters + search correct; invalid slug shows 404; back/forward work; Live Demo/Source buttons hidden for sample rows. **Prompt:** *"Phase 2: implement App Flow V4, V5, V8 and UI/UX Brief §5.5–5.7, 5.9. Use sample data only."*

Phase 3 — Cinematic intro and global motion

Tasks: `introMachine` reducer (+ unit tests), `IntroOverlay` (poster, muted autoplay, Skip, Play/Pause, Unmute, captions, timeout, error and autoplay-blocked fallbacks), transition choreography from UI/UX Brief §5.2, reveal stagger, Replay Intro in footer/menu, `pf:introSkip` with safe storage; Reveal/Stagger/TextReveal components, hover lift, image zoom, page transitions, reduced-motion handling; built-in generated poster + short placeholder video option so the whole flow is testable without the real video. **Exit criteria:** ended + skip simultaneously produce one transition; failing video never blocks the site; hidden overlay controls not focusable; reduced motion = simple fade; no white flash. **Prompt:** *"Phase 3: implement TRD §7–§8 and App Flow V1–V2 exactly. Include tests for all state transitions."*

Phase 4 — Business features

Tasks: services, packages (flag), FAQ accordion, testimonials slider (auto-hides when empty), contact form (all states in App Flow V7), `submit-inquiry` and `track-event` Edge Functions, honeypot, Turnstile, rate limit, email notification; analytics events wiring; WhatsApp/email/copy buttons. **Exit criteria:** with real keys: success only after server ok, failure + retry works, rate limit works; without keys: clear "Configuration needed" message and honest demo-mode text. **Prompt:** *"Phase 4: implement PRD §6.6–6.7, TRD §10 and Backend Schema §9 and §12. Never show success unless the server returned ok:true."*

Phase 5 — Admin core

Tasks: `/admin` guard, login, forgot password, optional TOTP, idle timeout; admin shell and design; dashboard (KPIs, launch readiness, sample counter); Projects CRUD with Tiptap, media picker, drag reorder, autosave, preview, publish/schedule, SEO fields; Media Library; Settings; Navigation & Sections; Intro Manager; Inbox. **Exit criteria:** create project

in admin → publish → appears on public site; draft not visible publicly; editor cannot reach owner-only pages; all RLS tests in Backend Schema §17 pass (in Supabase mode). **Prompt:** *"Phase 5: implement Admin flows A1–A8 and UI/UX Brief §7. Demo mode admin must work with IndexedDB and be disabled in production builds."*

Phase 6 — Admin advanced

Tasks: Categories/Tags, Skills & Tools, Experience/Education/Certifications/Achievements, Services/Packages/FAQ, Testimonials approval, Blog, Legal pages, SEO manager, CV manager, Analytics dashboard, **Sample Data Manager** (counts, replace shortcuts, type-to-confirm purge), revisions with restore, audit log viewer, Users & roles, Backup export/import with diff, danger zone, conflict handling on concurrent edits. **Exit criteria:** every public text/image/link is editable from admin; purge leaves real data untouched; export → import round-trip restores content. **Prompt:** *"Phase 6: implement PRD §8 (AD-10 to AD-53) and App Flow A9–A13."*

Phase 7 — Personalization and extras

Tasks: command palette, blog public pages, language switcher (only languages marked complete), RTL support, custom cursor and magnetic buttons (flagged, off on touch), optional lazy-loaded 3D hero (flag default off), social share. **Exit criteria:** all features obey flags; disabling a flag removes the feature cleanly.

Phase 8 — Hardening and launch

Tasks: pre-render public routes, sitemap/robots/JSON-LD/OG; CSP and security headers; Lighthouse and axe fixes; full Playwright suite; real data entry through admin; delete sample data; deploy; post-launch verification. **Exit criteria:** launch checklist (§6) fully green.

5. Acceptance Checklist per Phase (summary)

- [ ] No console errors or warnings introduced.
- [ ] Works at 360, 390, 768, 1024, 1440, 1920 px and landscape phone.
- [ ] Keyboard-only usage works; focus visible; dialogs trap and return focus.
- [ ] Loading, empty and error states exist for every data view.
- [ ] No placeholder button, dead link or fake statistic.

- [ ] Reduced motion and JS-animation failure leave content visible.
- [ ] New features covered by tests listed in TRD §17.
- [ ] Builder's report states: *works / needs configuration (exact variable) / known issues*.

6. Launch Checklist

Content: real profile and portrait; intro video + poster + captions uploaded (or intro disabled); CV PDF uploaded; at least 3 real projects with alt text; skills/tools, experience, services real; testimonials only with permission; privacy policy reviewed; sample data count = 0. **Config:** Supabase project, migrations applied, Owner created, Edge Functions deployed; Resend key + verified sender; Turnstile keys; `VITE_SITE_URL`; no demo credentials in production. **Quality:** Lighthouse mobile targets met; axe has no serious issues; contact form sends a real email; intro tested on iOS Safari, Chrome Android, desktop Chrome/Firefox/Safari; 404 and error pages tested; `noindex` on admin and staging. **SEO:** titles/descriptions per page, OG image, sitemap submitted, structured data validated, canonical URLs. **Safety:** RLS tests pass; no secrets in bundle; backup export taken.

7. Owner Runbook — From Dummy Data to Real Data

1. Log in to `/admin` → Dashboard → follow **Launch Readiness**.
2. **Settings:** name, title, bios, portrait, contact email, WhatsApp, socials, availability.
3. **Intro:** upload your AI video (MP4 H.264, ⩽ 15 MB recommended), poster, optional mobile crop, captions; set focal point; use *Test* buttons.
4. **CV:** upload PDF.
5. **Projects:** for each real project, edit a sample project or create a new one; fill role, process, results (only true facts), upload images with alt text; publish. Delete unused sample projects.
6. **Skills/Experience/Services/FAQ:** replace sample rows with your real information.
7. **Testimonials:** add only approved quotes; tick *permission confirmed*; otherwise leave empty (the section hides itself).
8. **SEO:** set page titles/descriptions and OG image.
9. **Sample Data** screen: confirm remaining count, then **Delete all sample data**.
10. Send yourself a test inquiry; confirm email arrives and the inbox entry exists.
11. Export a backup JSON, then connect your domain.

8. Asset Collection Checklist (prepare in parallel)

Asset	Specification	Status
AI intro video	16:9 MP4 H.264, 20–30 s, clear audio, face in central safe area; optional 9:16 / 4:5 mobile crop	☐
Poster image	WebP, ⩽ 150 KB, same framing as first frame	☐
Captions	`.vtt` English (and Roman Urdu if used)	☐
Portrait	4:5, ⩾ 1200 px wide	☐
Project media	Cover 16:10 + 3–6 screenshots each, ⩾ 1600 px wide	☐
CV	PDF ⩽ 10 MB	☐
Texts	Bio (short + long), value proposition, service descriptions	☐
Links	Email, WhatsApp, social profile URLs, booking link	☐
Accounts	Supabase, Vercel, GitHub, Resend, Cloudflare Turnstile, domain	☐

Intro scripts and the AI video generation prompt are in the original spec file (sections 26 and 27).

9. Risks and Contingencies

Risk	Contingency
AI Studio cannot connect Supabase from the preview	Keep demo mode for design/testing; deploy to Vercel and connect Supabase there
Builder output too large in one go	Split phase prompts per feature (e.g. Phase 5 into 5a auth/shell, 5b projects, 5c media/settings)
Edge Functions not available	Use a Vercel serverless function (`/api/submit-inquiry`) with the same contract (Backend Schema §12)

Risk	Contingency
Pre-rendering unsupported	Ship SPA with per-route meta, then migrate the same components to Next.js if SEO needs more
Video too heavy on mobile	Provide mobile source; lower bitrate; allow disabling intro from admin
Scope creep	Anything not in the PRD goes to a “Later” list; optional features stay behind flags

10. Maintenance After Launch

- Monthly: dependency updates, Lighthouse run, test the contact form end-to-end, review inbox and spam.
- Quarterly: refresh projects and CV, review analytics, verify all external links, export backup.
- Always: after any change, test intro skip/replay and the contact form before announcing updates.